CHAPTER 2

DISCRETE PROBABILITY DISTRIBUTIONS

CHAPTER ESSENCE AT A GLANCE:
Master Bernoulli, Binomial, and Poisson probability distributions, their parameter derivations (mean, variance), recurrence relations, and conditions for Poisson approximation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Bernoulli Distribution:** Models a single trial with two dichotomous outcomes (Success with probability p, Failure with probability q = 1 - p); Mean = p, Variance = pq.

• **Binomial Distribution B(n, p):** Models number of successes x in n independent repeated trials with constant probability p; PMF P(X=x) = nCx * p^x * q^(n-x); Mean = np, Variance = npq (Mean > Variance).

• **Poisson Distribution P(lambda):** Limiting form of Binomial distribution as n tends to infinity, p tends to 0, such that lambda = np remains finite; PMF P(X=x) = [e^(-lambda) * $lambda^x$] divided by x!; Mean = Variance = lambda.

• **Fitting of Distributions:** Estimating parameters from observed frequency data (p = Mean divided by n for Binomial; lambda = Mean for Poisson), calculating theoretical probabilities, and multiplying by N to obtain expected frequencies.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad \text{for } x = 0, 1, \ldots, n$$

Binomial Distribution Probability Mass Function

$$\text{Binomial Moments} \iff \text{Mean} = np, \quad \text{Variance} = npq \to \text{Mean} > \text{Variance}$$

Parametric Invariant of Binomial Distribution

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \ldots$$

Poisson Law of Rare Events Probability Mass Function

$$\text{Poisson Invariant} \iff \text{Mean} = \text{Variance} = \lambda$$

Equidispersion Identity of Poisson Distribution

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Binomial distribution, Mean is ALWAYS strictly greater than Variance (np > npq because q < 1); if a problem states Mean = 4 and Variance = 5, it is impossible.

• **Alert:** In Poisson distribution, Mean and Variance are strictly equal (Mean = Variance = lambda); this equidispersion is the diagnostic test for Poisson data.

• **Alert:** Poisson is applicable to rare events: radioactive decay, printing errors per page, road accidents per month, incoming phone calls per minute.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the conditions under which a Binomial distribution tends to a Poisson distribution. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Three Limiting Conditions:** (1) Number of trials n is infinitely large (n tends to infinity); (2) Constant probability of success p is extremely small (p tends to 0); (3) The product np = lambda remains finite and constant. **[2 Marks]**

2. For a Binomial distribution, Mean is 6 and Variance is 4. Find the parameters n and p and calculate P(X = 0). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Given:** Mean = np = 6; Variance = npq = 4. **[0.5 Mark]**

• **Find p and q:** Divide Variance by Mean: q = (npq) divided by (np) = 4 divided by 6 = 2 divided by 3. Then p = 1 - q = 1 - (2 divided by 3) = 1 divided by 3. **[1 Mark]**

• **Find n:** n * p = 6 implies n * (1 divided by 3) = 6, so n = 18. **[0.5 Mark]**

• **Calculate P(X = 0):** $P(X=0) = q^n = (2 \text{ divided by } 3)^{18}$. **[1 Mark]**

3. Six coins are tossed simultaneously. Find the probability of getting: (a) Exactly 4 heads, (b) At least 4 heads. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Parameters:** n = 6, p = 1 divided by 2 (fair coin), q = 1 - (1 divided by 2) = 1 divided by 2. Total outcomes = $2^6 = 64$. **[1 Mark]**

• **(a) Exactly 4 Heads P(X = 4):** $6C4 * (1 \text{ divided by } 2)^4 * (1 \text{ divided by } 2)^2 = 15 * (1 \text{ divided by } 64) = 15 \text{ divided by } 64 = 0.234$. **[1.5 Marks]**

• **(b) At Least 4 Heads P(X >= 4):** P(X=4) + P(X=5) + P(X=6) = [6C4 + 6C5 + 6C6] divided by 64 = (15 + 6 + 1) divided by 64 = 22 divided by 64 = 11 divided by 32 = 0.344. **[1.5 Marks]**

4. A manufacturer of pins finds that 2 percent of pins produced are defective. Using Poisson distribution, find the probability that a box of 100 pins contains: (a) No defective pins, (b) Exactly 1 defective pin (given e^(-2) = 0.1353). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Parameters:** n = 100, p = 0.02. Parameter lambda = np = 100 * 0.02 = 2. **[1 Mark]**

• **(a) No Defective P(X = 0):** $[e^{(-2)} * 2^0]$ divided by 0! = e^(-2) * 1 divided by 1 = 0.1353. **[1 Mark]**

• **(b) Exactly 1 Defective P(X = 1):** $[e^{(-2)} * 2^1]$ divided by 1! = 0.1353 × 2 = 0.2706. **[1 Mark]**

5. Can a Binomial distribution have Mean = 5 and Variance = 7? Justify your answer. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Conclusion:** No, this is mathematically impossible. **[1 Mark]**

• **Justification:** In a Binomial distribution, Variance = npq = Mean * q. Since q = 1 - p is a probability less than 1, Variance must be strictly less than Mean (npq < np). Here Variance (7) > Mean (5), which violates binomial theory. **[1 Mark]**

6. State the mode of a Poisson distribution when: (a) lambda is not an integer, (b) lambda is an integer. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) lambda is Non-Integer:** The distribution is unimodal, and Mode is the integral part of lambda [lambda]. **[1 Mark]**

• **(b) lambda is an Integer:** The distribution is bimodal with two modal values at lambda - 1 and lambda. **[1 Mark]**